Radio Direction Finding

Quick briefing

Radio direction finding estimates the direction from which an RF signal arrives. Use that bearing to support a decision while preserving the difference between a direction and a position.

Know the facts

Direction of arrival · DOA is the azimuthal arrival angle relative to the RDF system.

Phase comparison · Phase comparison, including the interferometric example, compares phase differences across antennas or antenna elements.

Time difference of arrival · TDOA compares when a signal arrives at spatially separated antennas.

Sources and applications · Possible sources include communications, radar and distress beacons. The lesson discusses navigation assistance, search and rescue, and spectrum monitoring.

Limits of a bearing · A single direction can inform coordination while distance, precise location and transmitter identity still require other evidence.

Understand the reasoning

State the reference for a direction before a recipient uses it. A bearing given relative to a system needs context to support a shared understanding.

A bearing can guide what a rescue team verifies next. Keep a possible source association tentative until independent exercise information supports it.

Philippine Army classroom exercise

A Philippine Army rescue tabletop presents a single synthetic bearing associated with a possible distress beacon. The exercise map shows two landing points in that direction, and the team can verify only one immediately.

Decision · What should the team relay, and how should the coordinator use it?

Worked analysis · Relay the measured direction with its reference and keep both landing points open as possibilities. Use independent exercise reports and current access information to prioritize a check. The bearing narrows the discussion, but it cannot select one location or identify the beacon by itself.

Check yourself

What does TDOA compare? · Differences in the time at which the same signal arrives at separated antennas.

Does one bearing establish an exact location? · No. It describes a direction, while a precise location requires additional evidence.

Course source

Radio Direction Finder (Revised)